

Sample House, Ground Floor Dwelling

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Fire Safety Strategy (Brannkonsept)

1. Scope and basis

This strategy covers the single-storey dwelling with flat roof modelled in the IFC file Ifc4_SampleHouse.ifc. It establishes the passive fire protection requirements for the building envelope, internal separations and the ground floor construction so the model can be checked for compliance with TEK17 chapter 11 prior to submission.

Risk class 4, fire class 1 (dwelling, one storey, one fire compartment). Classification notation follows NS-EN 13501-2. Requirements are stated per construction type as referenced in the model. GlobalIds are not used in this report; element types are identified by their Reference property.

Assumptions: no sprinkler installation; escape is direct to open air from the entrance hall; the roof-level "Simple floor" slab is a plant access deck and is not an occupied storey.

2. Construction types in scope

Model reference	Element	Location
Wall-Ext_102Bwk-75Ins-100LBlk-12P	External cavity wall	Ground Floor, three runs
Wall-Partn_12P-70MStd-12P	Internal stud partition	Ground Floor, two runs
Doors_IntSgl 810x2110mm	Internal single-leaf door	Ground Floor, two leaves
Doors_ExtDbI_Flush 1810x2110mm	External double door	Ground Floor, one set
Floor-Grnd-Susp_65Scr-80Ins-100Blk-75PC	Suspended ground floor slab	Ground Floor

3. Fire resistance requirements

3.1 External walls

The external cavity walls (Wall-Ext_102Bwk-75Ins-100LBlk-12P) shall achieve fire resistance class EI 30 as separating construction towards the boundary, in accordance with TEK17 §11-6 for fire class 1. The outer leaf shall be of reaction to fire class A2-s1,d0. The walls are non-load-bearing; the load path is via the internal blockwork leaf and the slab. No REI requirement applies.

3.2 Internal partitions

The stud partitions (Wall-Partn_12P-70MStd-12P) separating the bedroom from the entrance hall and living room form the escape route enclosure and shall achieve EI 30. Board linings shall be class K2 10 / B-s1,d0 minimum. These partitions are non-load-bearing and extend to the underside of the slab above; the head detail shall be fire-stopped to maintain the EI 30 line.

3.3 Internal doors

Internal doors (Doors_IntSgl 810x2110mm) in the EI 30 partitions shall be fire doors of class EI 30-Sa with self-closing devices per NS-EN 14600. The door schedule shall record the fire rating on each leaf.

3.4 External door

The external double door (Doors_ExtDbl_Flush 1810x2110mm) is the final exit and is an external element. No fire resistance requirement applies to it; it shall open in the direction of escape and be fitted with escape hardware per NS-EN 179.

3.5 Ground floor

The suspended ground floor slab (Floor-Grnd-Susp_65Scr-80Ins-100Blk-75PC) is load-bearing construction and shall achieve REI 30 per TEK17 §11-4 for the main load-bearing system in fire class 1.

4. Summary of requirements

Element	Requirement	Reference
External cavity wall	EI 30, A2-s1,d0 outer leaf, non-load-bearing	§3.1
Internal partition	EI 30, extends to structure	§3.2
Internal door	EI 30-Sa, self-closing	§3.3
External door	External element, no fire rating, escape hardware	§3.4
Ground floor slab	REI 30, load-bearing	§3.5

5. References

- TEK17 Byggeteknisk forskrift, kapittel 11 Sikkerhet ved brann
- NS-EN 13501-2 Fire classification of construction products, part 2
- NS-EN 14600 Doorsets and openable windows with fire resisting characteristics
- NS-EN 179 Emergency exit devices